Wenbo Zhu

PhD Researcher | Intelligent Systems · Robot Learning and Planning · Computer Architecture

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EDUCATION

University of Southern California, Los Angeles, CA **2024 – Present**
Ph.D. in Electrical and Computer Engineering | GPA: 4.0 | Advisor: Prof. Christopher Torng
Potential Thesis Title: Systems and Architectures for Efficient Robot Learning and Planning

Tsinghua University, Beijing, China **Jun 2024**
Bachelor of Engineering in Precision Instrument | GPA: 3.8

RESEARCH EXPERIENCE

Efficient Physical AI & Large-Scale Silicon Prototyping **Aug 2024 – Present**
USC | Acorn Research Group | Ph.D. Researcher | Advisor: Prof. Christopher Torng

- Investigate systems and architectures for efficient embodied AI, including vision-language-action and world models.
- Profiled CEM-based planning across three JEPA world models; validated W8A8 quantization with negligible impact on planning quality.
- Co-designed reference-forest sampling and a differential bit-sparse accelerator, achieving up to 1.80× speedup and 52% energy savings over equal-area dense hardware.
  (submitted to MLSys 2027, also **advised by Prof. Zishen Wan**)
- Contributed to mflowgen ASIC automation and multi-project wafer integration for rapid silicon prototyping, leading to the co-authored Chipstitch paper. (ICCAD 2026, ASPLOS 2027 in flight)
- Collaborated on PowerFlow-DNN, reporting up to 37% lower energy while meeting real-time inference deadlines. (ISLPED 2026, Best Paper Nomination)

Neural Rendering Acceleration and ASIC Co-Design **Mar 2022 – Mar 2023**
Georgia Tech | EIC Lab | Research Intern | Advisor: Prof. Yingyan (Celine) Lin

- Profiled TensoRF and Instant-NGP to identify training and inference bottlenecks in neural radiance fields. (ICCAD 2022)
- Modified Instant-NGP CUDA kernels and training/rendering pipelines to evaluate model compression and systems optimizations with near-lossless quality.
- Contributed workload characterization, CUDA reference changes, and architecture design informing a NeRF accelerator ASIC tapeout; co-authored papers at ISCA 2023 & MICRO 2024 (Best Paper Award)

Commercial Compute-in-SRAM Workload Optimization **Jul 2023 – July 2024**
Cornell | Computer Systems Lab | Intern / Collaborator | Advisor: Prof. Zhiru Zhang

- Characterized the GSI Gemini APU across seven Phoenix applications, matrix multiplication, and retrieval-augmented generation workloads.
- Used Roofline and analytical modeling to identify bottlenecks across a four-level memory hierarchy and 32K-element vector registers.
- Mapped kernels using communication-aware reductions, coalesced DMA, and broadcast-friendly layouts; the work reports 54.4×–117.9× lower retrieval energy than an NVIDIA A6000, published at MICRO 2025.

TECHNICAL SKILLS

Programming: Python, C/C++, CUDA, Triton, SystemVerilog/Verilog, Assembly

AI and ML Systems: PyTorch; PPO/DPO implementations; world models; robot simulation; language-model and distributed training; Triton kernels tuning

Systems and Architecture: Hardware benchmarking, performance profiling, Roofline modeling, compute-in-SRAM programming, hardware–software co-design

ASIC Tools: Cadence Innovus; Synopsys Design Compiler, VCS, Calibre, mflowgen; Experience with TSMC 28nm & 40nm

PUBLICATIONS

- **Candelta: Differential Sparsity Aware Planner-Hardware Co-Design for Accelerating World Model Planning**
  Wenbo Zhu, Paul Chen, Yuanhan Li, Zishen Wan, and Christopher Torng
  Submitted to ***MLSys, 2027***
- **A Compiler-Directed Polymorphic Silicon Query Architecture for Scalable and Cost-Efficient Accelerated Silicon Prototyping**
  Jeongeun Kim, Paul Yi-Chia Chen, Yuanhan Li, Sabrina Yarzada, **Wenbo Zhu**, Erin Coulon, Alena Novikova, Yujie Cai, Shriram Krishnamachari, Aidan Parris, and Christopher Torng
  Submitted to ***ASPLOS, 2027***
- **Chipstitch: A Scalable and Cost-Efficient Platform for Accelerated Silicon Prototyping**
  Claire Kim, Sabrina Yarzada, Paul Chen, Yuanhan Li, **Wenbo Zhu**, and Christopher Torng
  ICCAD, 2026
- **PowerFlow-DNN: Compiler-Directed Fine-Grained Power Orchestration for End-to-End Edge AI Inference**
  Paul Chen, Claire Kim, **Wenbo Zhu**, Yuanhan Li, Shunyao Huang, Chenjie Weng, and Christopher Torng
  ★ Best Paper Nomination
  ISLPED, 2026
- **Characterizing and Optimizing Realistic Workloads on a Commercial Compute-in-SRAM Device**
  Niansong Zhang, **Wenbo Zhu**, Courtney Golden, Dan Ilan, Hongzheng Chen, Christopher Batten, and Zhiru Zhang
  MICRO, 2025
- **Fusion-3D: Integrated Acceleration for Instant 3D Reconstruction and Real-Time Rendering**
  Sixu Li, Yang Zhao, Chaojian Li, Bowei Guo, Jingqun Zhang, **Wenbo Zhu**, Zhifan Ye, Cheng Wan, and Yingyan (Celine) Lin
  ★ Best Paper Award
  MICRO, 2024
- **Instant-3D: Instant Neural Radiance Field Training Towards On-Device AR/VR 3D Reconstruction**
  Sixu Li, Chaojian Li, **Wenbo Zhu**, Boyang Tony Yu, Yang Zhao, Cheng Wan, Haoran You, Huihong Shi, and Yingyan (Celine) Lin
  ISCA, 2023
- **RT-NeRF: Real-Time On-Device Neural Radiance Fields Towards Immersive AR/VR Rendering**
  Chong Li, Sixu Li, Yang Zhao, **Wenbo Zhu**, and Yingyan (Celine) Lin
  ICCAD, 2022

TEACHING EXPERIENCE

University of Southern California **Fall 2025**

Principal Teaching Assistant, EE 599 Complex Digital Design

- Led TSMC 28 nm technology bring-up and integration with group tools; extended mflowgen ASIC flows to support additional design choices for student chip projects. Organized several mini lectures on design tool and memory macro usage.
- Reviewed digital designs and supported synthesis, place-and-route, and tapeout preparation through recurring technical reviews.